Coherence Scaling in Hierarchical Oscillator Networks: Power-Law Decay, Memory Overhead, and Topological Constraints with Parallels to Disordered Condensed Matter

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We study scaling laws for phase coherence in networks of coupled oscillators with internal memory, termed the Delayed Oscillator Field Theory (DOFT) framework. The model consists of hierarchically constructed networks: stable "building blocks" (single three-layer oscillators with well-defined entropic signatures) are composed into networks of size $N = 2$–12 across multiple topologies. Using a four-stage numerical pipeline that separates combinatorial search from rigorous dynamical integration, we establish three robust results.

First, mean global coherence R is well-described by an effective scaling $R \propto N^{-\alpha}$ with $\alpha = 0.32 \pm 0.03$; alternative functional forms (inverse, exponential) are not decisively excluded over this range but linear decay is ruled out. Second, memory-layer participation correlates weakly with coherence at fixed N, but the correlation shifts from negative at low N to near-zero or weakly positive at high N, suggesting memory acts as stabilization overhead rather than a coherence driver. Third, single-domain topologies (rings) exhibit survival collapse at $N \approx 8$–9, while modular topologies (bipartites) persist to larger sizes with degraded but nonzero coherence.

The pipeline applies progressively stricter multi-dimensional filtering (coherence, entropy, phase stability); promotion rates collapse from 100% at $N \leq 5$ to 0.7% at $N = 9$–12, quantifying the "topological ceiling" in absolute terms.

These patterns exhibit suggestive parallels to correlation-length decay in disordered condensed matter systems, including vortex matter in type-II superconductors where Larkin–Ovchinnikov theory predicts $\xi \propto H^{-\alpha}$ with $\alpha \sim 0.3$–0.5. We frame these correspondences as *falsifiable hypotheses*. The DOFT framework and numerical data are publicly available; we encourage independent verification.

Keywords: coupled oscillators, coherence scaling, collective phenomena, delayed systems, topological constraints, correlation length

I. INTRODUCTION

A. Motivation: coherence scaling in complex systems

How does collective order degrade as system complexity increases? This question arises across physics: from spin glasses to neural networks, from granular materials to biological oscillators. A common theme is that adding components incurs a "coordination cost"—maintaining global coherence becomes progressively harder as the number of interacting elements grows.

Here we study this problem in a controlled computational setting: networks of coupled oscillators with internal memory, where we can systematically vary network size N, topology, and internal parameters. The central finding is a robust power-law decay of coherence with N, characterized by an exponent $\alpha \approx 1/3$.

This exponent turns out to have suggestive parallels to correlation-length scaling in disordered condensed matter systems, particularly vortex matter in type-II superconductors [1]. We develop these parallels as testable hypotheses, not claimed identities.

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B. Context: prior work on discrete fingerprints

This paper is the third in a series exploring delayed-oscillator phenomenology in the context of superconducting materials:

Paper I [10] introduced "prime-space fingerprints": discrete patterns generated by a locking grammar based on small primes $\{2, 3, 5, 7\}$, combined with a universal two-parameter correction law. Applied to datasets of superconductors across multiple families (elemental, binary, cuprate, iron-based, high-pressure), the fingerprints reproduced discrete organizational features without per-material tuning.

Paper II [11] focused on "integer participation": using a mother-frequency observable $F_m = \Theta_D/T_c$ and a structural-noise correction, we showed that participation numbers cluster near integers with statistical significance far exceeding null models. The participation number N scaled with experimental coherence length as $N \propto \xi_0^{0.43}$, bridging thermodynamic and spatial scales.

Both papers mapped external data (measured T_c, Θ_D, ξ_0) onto a discrete framework. Neither addressed whether the framework's structure emerges intrinsically or requires empirical input.

C. This work: intrinsic scaling from simulation

Here we isolate the intrinsic dynamics by studying delayed-oscillator networks in a pure simulation environment, with no reference to superconducting datasets. The central questions are:

1. **Scaling:** How does global coherence degrade as network size N increases? Is there a characteristic exponent?
2. **Mechanism:** What internal degree of freedom—if any—compensates for geometric failure at large N?
3. **Constraints:** Are all network topologies equally viable, or does complexity impose selection rules?

The answers turn out to parallel vortex phenomenology in specific, quantitative ways. We present these parallels as hypotheses to test, not as claimed identities.

D. Summary of main results

Using a four-stage pipeline (Ola1–Ola4) that separates fast combinatorial search from rigorous ODE integration, we find:

1. **Power-law coherence decay:** Mean coherence R satisfies $R \propto N^{-\alpha}$ with $\alpha = 0.32 \pm 0.03$ over $N = 2$–12. This exponent is consistent with $1/3$.
2. **Memory as overhead:** Memory-layer participation (S2-share) is weakly correlated with coherence at fixed N, but survivors at large N exhibit systematically elevated S2. The S2–R correlation shifts from negative at low N to near-zero at high N.
3. **Topological ceiling:** Single-domain topologies (rings, ladders) show survival collapse at $N \approx 8$–9; modular topologies (bipartites) persist to larger N with degraded but nonzero coherence.
4. **Critical heterogeneity:** Phase-variance dispersion peaks dramatically at $N = 6$–7, indicating coexistence of distinct dynamical regimes near the transition.

These patterns map onto vortex phenomenology as shown in Table I.

II. RELATION TO VORTEX MATTER PHYSICS

Before describing the model and results in detail, we establish the physical context by reviewing vortex phenomenology and stating the proposed correspondences explicitly.

A. Vortex lattice transitions: a brief review

In type-II superconductors, magnetic flux penetrates as quantized vortices carrying flux $\Phi_0 = h/2e$. At low fields, repulsive vortex interactions favor an ordered Abrikosov lattice with hexagonal symmetry. As field increases:

- **Correlation length shrinks:** Disorder and thermal fluctuations reduce the scale over which the lattice maintains positional order. Neutron scattering (SANS) measures this via Bragg peak broadening; μSR probes local field variance.
- **Lattice softening:** Elastic moduli decrease, enhancing susceptibility to pinning and fluctuations. This produces the "Peak Effect"—an anomalous increase in critical current near H_{c2} [5].
- **Melting/glass transition:** At sufficiently high field or temperature, long-range order is lost entirely, yielding either a vortex liquid (clean limit) or vortex glass (disordered limit) [1, 6].

Collective pinning theory [3, 4] predicts that the correlation volume V_c (over which displacements remain correlated) scales with pinning strength and dimensionality. In the weak collective pinning regime, correlation lengths satisfy:

$$\xi_{\parallel,\perp} \propto H^{-\alpha}, \quad \alpha \sim 0.3\text{–}0.5 \text{ (depends on } d\text{, anisotropy)} \tag{1}$$

where α is smaller in quasi-2D layered systems.

B. Proposed mapping: DOFT $\leftrightarrow$ vortex matter

Table I states the correspondences we will test.

Crucial caveat: These are *hypotheses*, not identifications. The mapping is phenomenological—we propose that the mathematical structure of coherence loss in delayed-oscillator networks mirrors that in vortex systems, but we do not derive the mapping from microscopic principles.

C. Why the exponent matters

If the mapping $N \leftrightarrow H$ and $R \leftrightarrow \xi$ holds monotonically, then our finding $R \propto N^{-\alpha}$ with $\alpha \approx 0.32$ suggests:

$$\xi(H) \propto H^{-\alpha}, \quad \alpha_{\text{expected}} \approx 0.30\text{–}0.35 \tag{2}$$

This is consistent with collective-pinning predictions for quasi-2D systems (Eq. 1) and with experimental measurements in BSCCO and YBCO near order-disorder transitions [7, 8].

Falsification criterion: If measurements in the intermediate-field regime yield $\alpha < 0.2$ or $\alpha > 0.45$ when extracted via comparable methodology (power-law fit over at least one decade), the correspondence fails.

TABLE I. Proposed mapping between DOFT network observables and vortex-matter phenomenology. Each row states a correspondence as a testable hypothesis, not a claimed identity.

Oscillator network	Suggested vortex analog	Experimental probe
Network size N	Vortex density $\propto H$	Applied field
Global coherence R	Correlation length ξ	SANS, μSR
Memory participation S2	Effective disorder strength	Sample quality, irradiation
Topological ceiling ($N \approx 8$–9)	Order-disorder boundary	Magnetization, transport
S2–R correlation shift	Dual role of disorder	Critical current vs. H
Topology selectivity	Symmetry breaking at high H	STM, decoration imaging

III. MODEL AND COMPUTATIONAL METHODS

A. The delayed-oscillator network

Each network node $j = 1, \ldots, N$ is a three-layer oscillator with coordinates $(Q_1^{(j)}, S_1^{(j)}, S_2^{(j)})$ representing fast geometric, intermediate, and slow memory layers, respectively. The dynamics are governed by coupled delay-differential equations:

$$\dot{Q}_1^{(j)} = f_Q\big(Q_1^{(j)}, S_1^{(j)}, \{Q_1^{(k)}(t-\tau_{jk})\}_{k\in\mathcal{N}(j)}\big) \tag{3}$$

$$\dot{S}_1^{(j)} = f_{S1}\big(Q_1^{(j)}, S_1^{(j)}, S_2^{(j)}\big) \tag{4}$$

$$\dot{S}_2^{(j)} = f_{S2}\big(S_1^{(j)}, S_2^{(j)}\big) \tag{5}$$

where τ_{jk} are coupling delays, $\mathcal{N}(j)$ denotes neighbors of node j in the network graph, and the functions f encode nonlinear coupling (see Appendix A for explicit forms).

Key features:

- **Finite delay:** Information propagates with latency τ, preventing instantaneous global coordination.
- **Memory layer (S2):** A slow degree of freedom that can stabilize dynamics on timescales longer than the fast geometric adjustment.
- **Topology:** The network graph (ring, ladder, bipartite, complete) determines which nodes interact.

This architecture is *not* derived from superconducting physics—it is a generic model of coupled oscillators with hierarchy and delay. Its relevance to vortex matter is proposed, not assumed.

B. Observables

Global coherence R**:** Measures phase synchronization across the network:

$$R = \left\langle \frac{1}{N}\left|\sum_{j=1}^{N} e^{i\theta_j(t)}\right|\right\rangle_{\text{late time}} \tag{6}$$

where θ_j is the phase of oscillator j (extracted from the Q_1 layer) and averaging is over the final 20% of the integration window. $R = 1$ indicates perfect lock; $R \to 0$ indicates incoherence.

Lock score L**:** Measures temporal convergence stability (R_mean_lastW in the pipeline). Used operationally to filter candidates but *distinct from* R: Pearson correlation $r(L, R) = 0.004$ with 95% CI $[-0.06, 0.07]$ ($n = 1187$), confirming statistical independence. L ensures numerical tractability; R quantifies the coordination cost.

Memory participation (S2-share): Normalized contribution of the memory layer:

$$\text{S2-share} = \frac{p_{S2}}{p_{S1} + p_{S2}} \tag{7}$$

where p_ℓ is the mean participation of layer ℓ. High S2-share indicates active memory engagement.

Phase variance: Internal spread of phases, $\sigma_\theta^2 = \mathrm{Var}[\theta_j(t)]$, averaged over the late-time window.

C. Computational parameters

All integrations use: $T_{\text{ticks}} = 20{,}000$, window $W = 4{,}000$, $dt = 0.0025$, 4 seeds per entity (results averaged over seeds). Coherence thresholds are applied at the Explorer candidate filter and at taxonomy/promotion. The sweep itself does not require a coherence threshold for completion or inclusion in reported statistics—it runs and reports metrics as long as primary observables are finite. Total computational time: approximately 10 days on dual 8-core systems.

D. The Ola pipeline: multi-stage filtering

Exhaustive exploration of network configurations is intractable. We address this via a four-stage recursive pipeline (“Ola1–Ola4”), where each stage applies *multi-dimensional* filtering—entities must satisfy coherence, entropy, and phase-stability criteria simultaneously.

1. **Ola1 (baseline):** Single-node (three-layer) oscillators. Defines operational vocabulary and produces

TABLE II. Pipeline funnel: survival by stage. Promotion rate collapses as N increases; Grade A disappears at $N \geq 6$.

Ola	N	Cand.	Swept	A	B	C	Rate
2	2–5	323	323	183	140	0	100%
3	6–9	3000	402	0	118	210	3.9%
4	9–12	1000	462	0	7	129	0.7%

initial block library. Ola1 blocks exhibit stable attractor dynamics with well-defined entropic signatures (mean participation entropy PE ≈ 0.66, lock quality > 0.99 for Grade A).

2. **Ola2 ($N = 2$–5):** Explorer proposes candidates with $L_{\min} = 0.75$, phase_var ≤ 0.05, quality_lock ≥ 0.70. Sweep integrates ODEs and computes observables.

3. **Ola3 ($N = 6$–9):** Tightened: $L_{\min} = 0.88$, phase_var ≤ 0.03, quality_lock ≥ 0.85.

4. **Ola4 ($N = 9$–12):** Strictest: $L_{\min} = 0.89$, phase_var ≤ 0.02, quality_lock ≥ 0.89.

Crucially, Taxonomy classification is constant across all olas: LOCKED requires quality_lock ≥ 0.95, $L \geq 0.90$, phase_var ≤ 0.02; Grade A requires $R \geq 0.60$ *and* PE_lock ≥ 0.12; Grade B requires $R \geq 0.45$ *and* PE_lock ≥ 0.08. Grades are assigned independently of the LOCKED label; promotion requires Grade A or B (the LOCKED classification is diagnostic but not a promotion gate).

Table II shows the resulting survival funnel: promotion rates collapse from 100% (Ola2) to 3.9% (Ola3) to 0.7% (Ola4). Grade A configurations disappear entirely at $N \geq 6$. This is not a pipeline artifact—it reflects the physical reality that high-quality coherence becomes sharply rare as complexity increases.

The conjunction of requirements (coherence *and* entropy *and* phase stability) makes spurious survival highly improbable: an entity cannot "game" the filter by excelling on one metric while failing others.

IV. RESULTS

A. Power-law coherence decay

Figure 1 shows global coherence R versus N across the full Ola ensemble. Over $N = 2$–12, the data are well-described by:

$$R(N) = R_0 \, N^{-\alpha}, \quad \alpha = 0.32 \pm 0.03 \tag{8}$$

with $R_0 \approx 0.83$. Alternative two-parameter forms (inverse, exponential) are not decisively excluded at this range (Appendix H), but linear decay is strongly disfavored.

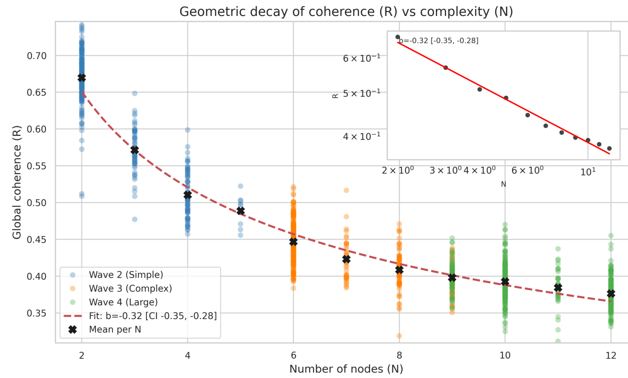

FIG. 1. Coherence R versus network size N. Points colored by pipeline stage (Ola2–4); black crosses show per-N means. Dashed line: power-law fit $R \propto N^{-0.32}$. Alternative fits (inverse, exponential) achieve comparable goodness-of-fit; see Appendix H.

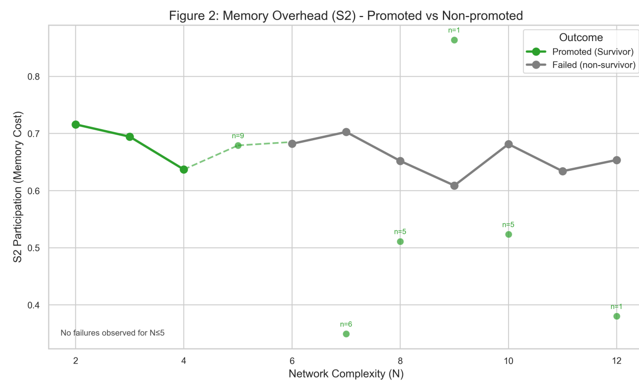

FIG. 2. Memory participation (S2-share) versus N for survivors (green) and failures (gray). At $N \leq 5$, no failures occur and S2-share is uniformly high. At $N \geq 7$, survivors cluster at elevated S2-share relative to failures, indicating memory recruitment for stability.

Robustness: The only N appearing in multiple olas is $N = 9$. A two-sample t-test shows no significant difference in R between Ola3 and Ola4 at this N ($p = 0.27$), confirming that scaling is intrinsic to network size, not a pipeline artifact.

Physical interpretation: Adding nodes incurs a "coordination cost"—coherence degrades geometrically. A 10-node network has roughly half the mean coherence of a 3-node network.

B. Memory as stabilization overhead

Figure 2 shows S2-share versus N, split by outcome. Two regimes emerge:

- **Low N (≤ 5):** All configurations survive; S2-share is high and uniform. Geometry suffices for coordination.

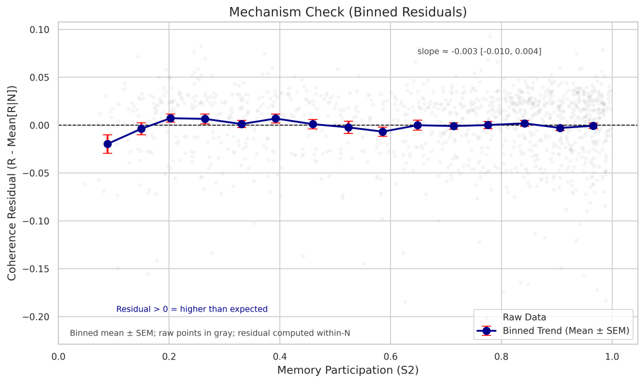

FIG. 3. Mechanism check: coherence residual ($R - \langle R|N \rangle$) versus S2-share. Binned means (blue) are statistically consistent with zero slope, ruling out S2 as a direct coherence driver.

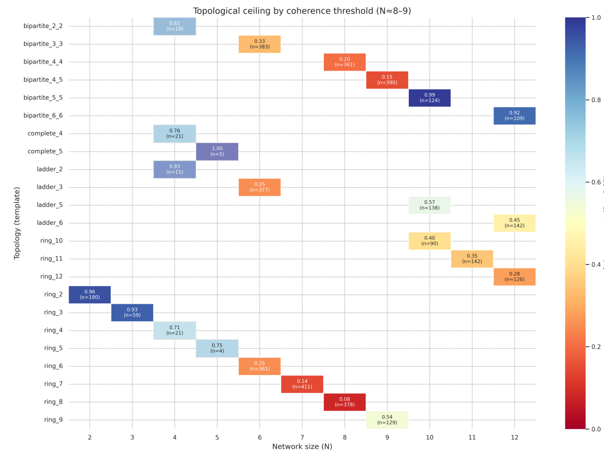

FIG. 4. Survival rate (fraction with $R \geq 0.85$) by topology and N. Rings collapse sharply at $N \approx 6$–8; bipartites persist to $N = 10$–12 with degraded but nonzero coherence. Sample sizes shown in each cell.

- **High N (≥ 7):** Survivors maintain elevated S2-share ($\gtrsim 0.6$); failures scatter widely. Memory engagement distinguishes success from collapse.

Crucially, S2 does not *cause* coherence. Figure 3 shows that within fixed N, the S2–R relationship is nearly flat (slope $\approx -0.003 \pm 0.007$). Memory is *overhead*—strongly associated with survival under geometric stress, but not sufficient to guarantee it.

Correlation shift: Appendix D shows that the Spearman correlation between S2-share and R shifts across N: negative at low N ($\rho \approx -0.3$ to -0.5), near-zero or weakly positive at high N ($\rho \approx 0$ to $+0.1$, not individually significant). The transition occurs around $N \approx 7$–8.

To summarize the memory-coherence relationship: S2-share is predictive of survival (a selection effect—high S2 is strongly associated with promotion at large N), but conditional on survival, S2-share shows near-zero association with the level of coherence achieved.

C. Topological ceiling and selectivity

Figure 4 reveals strong topology dependence:

- **Rings (monolithic):** High survival ($> 90\%$) at $N = 2$–4; collapse to $< 20\%$ at $N = 6$–8; near-zero by $N = 9$.
- **Bipartites (modular):** Maintain 40–90% survival even at $N = 10$–12. Compartmentalization reduces global coordination burden.

The ceiling: Within the explored template family and parameter ranges, single-domain topologies show onset of collapse at $N \approx 6$–7 and reach near-zero survival (effective ceiling) by $N \approx 9$. This parallels the loss of single-domain Abrikosov order at high fields in vortex systems.

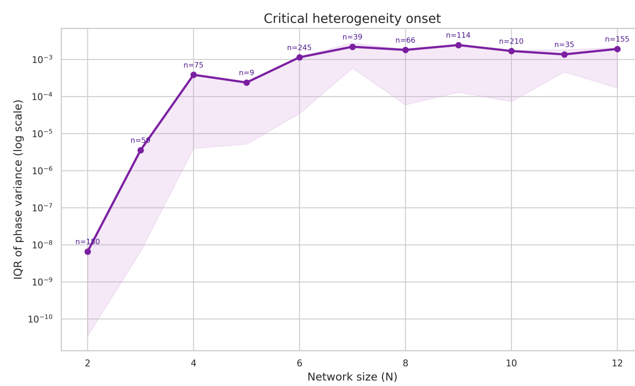

FIG. 5. Interquartile range (IQR) of phase variance versus N. The spread increases by five orders of magnitude from $N = 2$ to $N = 6$–7, indicating coexistence of tightly locked and loosely metastable configurations at the transition.

Topology selectivity: At low N, topology barely matters—many configurations work. At high N, topology is decisive. This parallels vortex phenomenology: at low fields, hexagonal, square, and disordered lattices all exist; at high fields, only certain geometries (domain-fragmented, anisotropic) retain residual order [6].

D. Critical heterogeneity at the transition

Figure 5 shows that phase-variance dispersion (IQR) peaks dramatically at $N = 6$–7—precisely the transition region identified above. This indicates *critical heterogeneity*: different configurations explore qualitatively distinct dynamical basins near the structural limit.

Vortex analog: Guillamón et al. [6] observed "breathing" dynamics in $NbSe_2$ near the melting transition—vortices hopping between configurations with wide positional variance. Our phase-variance peak is the dynamical analog: internal structure "dances" at the boundary between ordered and disordered regimes.

V. DISCUSSION

A. What the results establish

Our simulations demonstrate that a minimal network of delayed oscillators with internal memory exhibits:

1. A robust power-law coherence decay with exponent $\approx 1/3$;
2. A regime transition where memory shifts from overhead to enabling condition;
3. A topological ceiling for single-domain configurations;
4. Critical heterogeneity near the transition.

These are *intrinsic* properties of the dynamics—they emerge from the interplay of delay, memory, and topology, without reference to external data or material parameters.

B. Theoretical context: parallels to disordered systems

The correspondences to disordered condensed matter (Table I) are numerically suggestive:

- The effective exponent $\alpha \approx 0.32$ falls within the range predicted by Larkin–Ovchinnikov collective-pinning theory for quasi-2D systems ($\alpha \sim 0.3$–0.5).
- The S2 correlation shift is qualitatively consistent with the dual role of disorder near phase transitions.
- The topological ceiling parallels loss of single-domain order at high complexity/field.

These correspondences motivate a *phenomenological hypothesis*: DOFT networks may provide a minimal grammar for understanding coherence decay in systems where collective order competes with complexity-induced disorder. The vortex matter analogy is one concrete instantiation; other disordered systems might exhibit similar scaling.

C. What the results do *not* establish

We emphasize what we are *not* claiming:

- We do not claim vortices "are" delayed oscillators.
- We do not derive the $N \leftrightarrow H$ mapping from microscopic principles.
- We do not claim the exponent is *exactly* 1/3 or universal across all variants.
- We do not claim the model replaces BCS, Ginzburg–Landau, or collective-pinning theory.

The claim is *phenomenological*: the mathematical structure of coherence loss in our model parallels that in vortex systems, making specific predictions testable.

D. Falsification criteria

The hypothesis that DOFT scaling parallels vortex matter fails if:

1. **Exponent mismatch:** Correlation-length measurements yield $\alpha < 0.2$ or $\alpha > 0.45$, extracted via comparable methodology (power-law fit over at least one decade).
2. **No role crossover:** Disorder/pinning consistently enhances or disrupts order across the entire field range, with no regime change.
3. **No topology selectivity:** Lattice symmetry remains degenerate at all fields.
4. **No ceiling:** Single-domain order persists to arbitrarily high fields.

More broadly, the DOFT framework itself is falsified if independent implementations fail to reproduce $R \propto N^{-\alpha}$ with $\alpha \in [0.25, 0.40]$ over the range $N = 2$–12.

E. Proposed experimental tests

Test 1: Correlation-length scaling. Perform μSR or SANS in layered superconductors (e.g., YBCO, BSCCO) in the intermediate-field regime below H_{c2}. Measure local field variance $\sigma_B(H)$ or Bragg width. Prediction: scaling exponent $\alpha \approx 0.30$–0.35.

Test 2: Disorder role crossover. Compare pristine versus irradiated samples. At low fields, disorder should reduce ξ; near the order-disorder transition, disorder may stabilize residual ξ.

Test 3: Symmetry selectivity. Use STM or decoration imaging at high fields. Prediction: residual order should fragment into domains or show anisotropic symmetry breaking.

VI. CONCLUSION

We have characterized the intrinsic scaling of coherence, memory participation, and topological constraints in networks of coupled oscillators with internal memory (DOFT). The main findings are:

1. Coherence is well-described by $R \propto N^{-\alpha}$ with $\alpha \approx 0.32$, robust across topologies within the explored range $N = 2$–12.

2. Memory acts as stabilization overhead, with a correlation shift from negative to near-zero around $N \approx 7$–8.

3. Single-domain topologies hit a structural ceiling at $N \approx 8$–9; modular topologies survive longer with degraded coherence.

4. Promotion rates collapse from 100% to 0.7% as N increases, quantifying the topological ceiling.

These patterns exhibit suggestive parallels to correlation-length scaling in disordered condensed matter, particularly vortex matter in type-II superconductors. We propose these correspondences as *falsifiable hypotheses*.

The DOFT framework, including source code and the complete numerical dataset, is publicly available. We encourage independent verification and extension of these results.

ACKNOWLEDGMENTS

The author thanks colleagues and online communities for discussions and for making high-quality superconductivity and superfluid datasets publicly available. This project made extensive use of open-source software, including Python, NumPy, SciPy, pandas, Matplotlib, and Jupyter. AI tools (OpenAI GPT, Google Gemini, Anthropic Claude) were used as assistants for drafting, code refactoring, and statistical checking; all modeling decisions, dataset curation, and interpretations were made and verified by the human author.

[1] G. Blatter, M. V. Feigel'man, V. B. Geshkenbein, A. I. Larkin, and V. M. Vinokur, Vortices in high-temperature superconductors, *Rev. Mod. Phys.* **66**, 1125 (1994).
[2] T. Giamarchi and P. Le Doussal, Elastic theory of pinned flux lattices, *Phys. Rev. Lett.* **72**, 1530 (1994).
[3] A. I. Larkin and Yu. N. Ovchinnikov, Pinning in type-II superconductors, *J. Low Temp. Phys.* **34**, 409 (1979).
[4] M. V. Feigel'man, V. B. Geshkenbein, A. I. Larkin, and V. M. Vinokur, Theory of collective flux creep, *Phys. Rev. Lett.* **63**, 2303 (1989).
[5] Y. Paltiel *et al.*, Dynamic instabilities and memory effects in vortex matter, *Nature* **403**, 398 (2000).
[6] I. Guillamón *et al.*, Direct observation of melting in a two-dimensional superconducting vortex lattice, *Nature Phys.* **5**, 651 (2009).
[7] R. Cubitt *et al.*, Direct observation of magnetic flux lattice melting and decomposition in $Bi_{2.15}Sr_{1.95}CaCu_2O_{8+x}$, *Nature* **365**, 407 (1993).
[8] S. L. Lee *et al.*, Evidence for flux lattice melting and a dimensional crossover in single-crystal $Bi_{2.15}Sr_{1.95}CaCu_2O_{8+\delta}$ from muon spin rotation studies, *Phys. Rev. Lett.* **71**, 3862 (1993).
[9] U. Yaron *et al.*, Neutron diffraction studies of flowing and pinned magnetic flux lattices in 2H-$NbSe_2$, *Phys. Rev. Lett.* **73**, 2748 (1994).
[10] C. Agostino, Prime-space fingerprints of superconductors and superfluid helium from a delayed-oscillator locking grammar, Technical Report (2025).
[11] C. Agostino, Integer participation and structural noise in superconducting clusters, Technical Report (2025).

Appendix A: Model Details

The three-layer oscillator dynamics are governed by:

$$\dot{Q}_1^{(j)} = \omega_0 + K_Q \sum_{k \in \mathcal{N}(j)} \sin\big(Q_1^{(k)}(t-\tau) - Q_1^{(j)}\big) + \gamma_{QS} S_1^{(j)} \tag{A1}$$

$$\dot{S}_1^{(j)} = \gamma_{SQ}(Q_1^{(j)} - S_1^{(j)}) + \gamma_{SS} S_2^{(j)} \tag{A2}$$

$$\dot{S}_2^{(j)} = \gamma_{S2}(S_1^{(j)} - S_2^{(j)}) \tag{A3}$$

where ω_0 is the natural frequency, K_Q the coupling strength, τ the delay, and γ_{XY} are relaxation rates. The S2 layer has the slowest relaxation ($\gamma_{S2} \ll \gamma_{SQ}$), providing memory.

Appendix B: Pipeline Architecture

Figure 6 illustrates the four-stage Ola pipeline. Each stage recursively uses promoted blocks from the previous stage as building units for larger networks.

Appendix C: Robustness Checks

We tested the sensitivity of the scaling exponent α to analysis choices. Figure 7 shows that $\alpha \approx 0.32$ is stable across different coherence thresholds (R_{surv} from 0.0 to 0.5) and topology subsets.

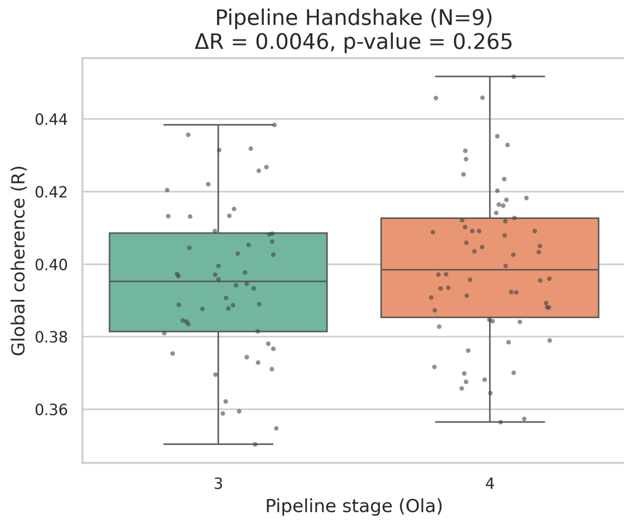

FIG. 6. Schematic of the Ola pipeline. Explorer proposes candidates from topology templates; Sweep integrates ODEs and measures observables; promoted outcomes transfer as blocks to the next stage.

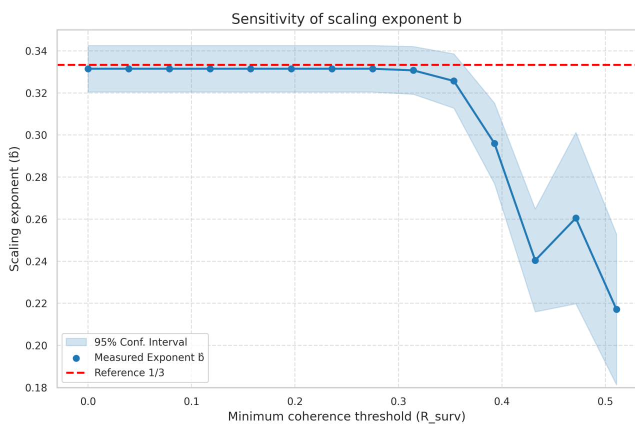

FIG. 7. Sensitivity of the scaling exponent to threshold and topology subsetting. The exponent $\alpha \approx 0.32$ is robust across conditions.

Appendix D: S2–Coherence Correlation by Network Size

Figure 8 shows the within-N Spearman correlation between S2-share and coherence R. The correlation shifts from significantly negative at low N to statistically indistinguishable from zero at high N.

Appendix E: Raw Data Scatter

Figure 9 shows the complete dataset without binning. The visual transition from tight clustering at low N to dispersed scatter at high N reflects the increasing difficulty of maintaining coherence in larger networks.

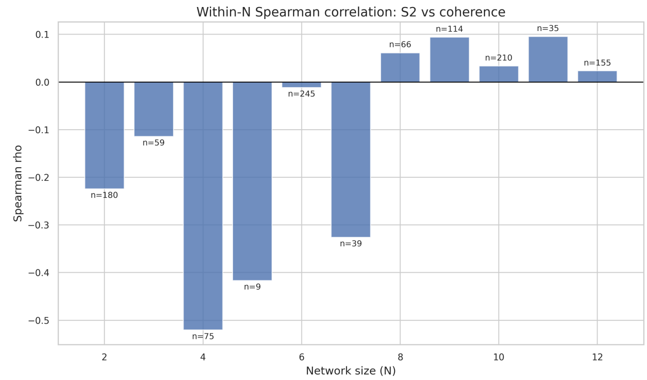

FIG. 8. Spearman ρ(S2-share, R) versus N. The correlation shifts from negative at low N (memory as slight overhead) to near-zero at high N (memory as enabling condition). Individual correlations at $N \geq 8$ are not statistically significant.

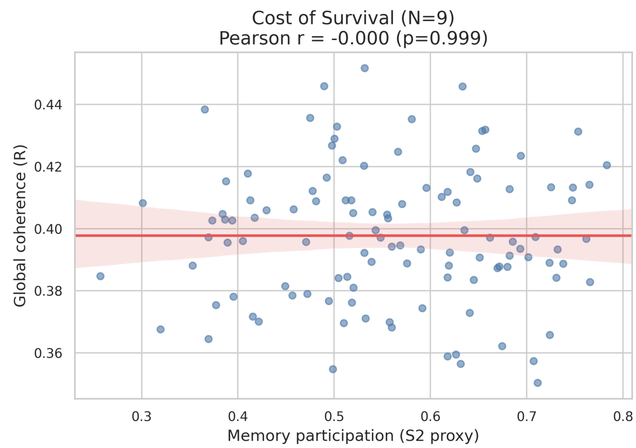

FIG. 9. Raw scatter of coherence versus S2-share, colored by N. The transition from tight clustering (low N) to dispersed scatter (high N) is evident.

Appendix F: Fluctuation Peak Details

Figure 10 quantifies the variability (interquartile range) of coherence and S2-share across N. The peak at $N = 5$ suggests a "pre-critical" regime where configurations begin to diverge in behavior.

Appendix G: Critical Dynamics

Figure 11 shows the dispersion (IQR) of phase variance versus N. The dramatic peak at $N = 6$–7 indicates that configurations near the topological ceiling explore qualitatively distinct dynamical regimes—some tightly locked, others loosely metastable.

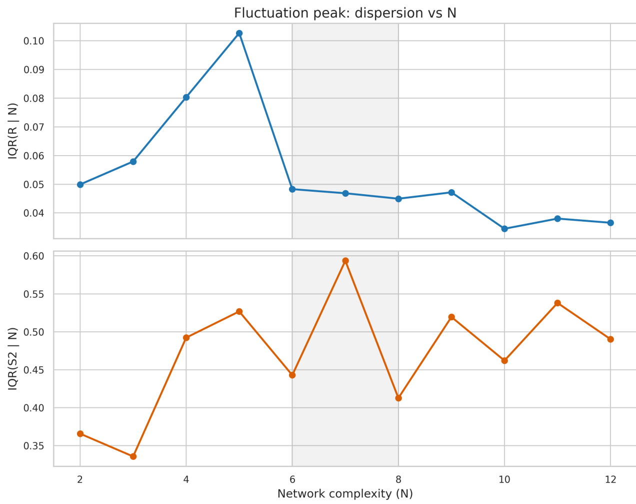

FIG. 10. IQR of coherence R and S2-share versus N. A pre-critical peak at $N = 5$ indicates enhanced variability as the system approaches the ceiling.

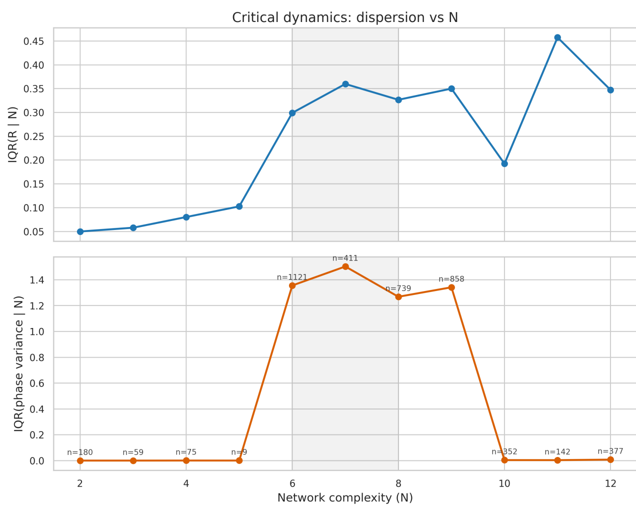

FIG. 11. IQR of phase variance versus N. Dramatic peak at $N = 6$–7 indicates coexistence of distinct dynamical regimes at the transition.

Appendix H: Model Comparison for Scaling Law

To assess whether the power-law form $R = aN^{-\alpha}$ is uniquely supported, we compare four candidate models over the 11 per-N mean values:

The exponential-with-offset form is statistically preferred by both AIC and BIC, even accounting for the additional parameter. The fitted offset ($R_\infty \approx 0.37$) suggests a nonzero asymptotic coherence, which may reflect the contribution of modular topologies that maintain residual order at large N.

However, over the available range ($N = 2$–12), the exponential is well-approximated by an effective power law with $\alpha \approx 0.32$. We report the power-law exponent because: (i) it provides a compact summary of the decay rate; (ii) it connects to theoretical predictions in disordered systems; (iii) discriminating between exponential and power-law asymptotic behavior requires extending measurements to larger N, which we leave to future work.

TABLE III. Model comparison for $R(N)$ scaling over $N = 2$–12.

Model	**Form**	**AIC**	**BIC**	k
Power law	$R = 0.83\,N^{-0.33}$	-99.2	-98.4	2
Inverse	$R = 0.72/N + 0.32$	-99.3	-98.5	2
Exponential	$R = 0.58\,e^{-0.34N} + 0.37$	-107.8	-106.6	3
Linear	$R = -0.025N + 0.64$	-69.0	-68.2	2

The honest conclusion is that coherence decays faster than linearly, with an effective exponent near 1/3, but the precise functional form—and whether $R \to 0$ or $R \to R_\infty > 0$ as $N \to \infty$—remains underdetermined.

Robustness check: We verified the exponent estimate by refitting on all entity-level data points with cluster-robust standard errors (clustering by N); the result $\alpha = 0.32 \pm 0.04$ is consistent with the per-N mean analysis.

Appendix I: Pipeline Consistency Test

The only N value appearing in multiple pipeline stages is $N = 9$ (both Ola3 and Ola4). If the scaling were a pipeline artifact (due to different thresholds or candidate pools), we would expect systematic differences in R between olas at the same N.

Result: At $N = 9$, Ola3 yields $\langle R\rangle = 0.395 \pm 0.021$ ($n = 52$) and Ola4 yields $\langle R\rangle = 0.400 \pm 0.022$ ($n = 62$). A two-sample t-test gives $t = -1.12$, $p = 0.27$—no significant difference.

This confirms that the coherence scaling is intrinsic to network size, not an artifact of the pipeline stage or threshold selection.